

Measure Theory  
Jan-May 2026    Anoop V. P.

## Introduction: From Riemann to Lebesgue

Measure theory provides a systematic way to assign numerical values to sets and functions. Specifically, we assign a **measure** to a measurable set and an **integral value** to an integrable function. While Riemann integration is a staple of introductory calculus, Lebesgue integration theory represents a significant generalization and completion of those ideas. Lebesgue's framework allows us to assign numbers to a much broader collection of sets and functions than is possible within the Riemann system.

### The Shift from Finite to Countable Operations

The fundamental difference between these two theories can be summarized by their relationship with infinity:

- **Riemann Integration** is centered on **finiteness**. It relies on approximations of a finite nature, such as summing the areas of finitely many rectangles to find the area of a bounded subset of  $\mathbb{R}^2$ . This theory is robust when dealing with finite operations on sets  $A_1, \dots, A_n$  or functions  $f_1, \dots, f_n$ .
- **Lebesgue Integration** is centered on **countable infiniteness**. Riemann integration often fails to behave consistently when faced with operations of a countably infinite nature, such as the union  $\bigcup_{n=1}^{\infty} A_n$ , the sum  $\sum_{n=1}^{\infty} f_n$ , or the limit  $\lim_{n \rightarrow \infty} f_n$ . Lebesgue integration was developed to rectify these disadvantages.

### Methodological and Structural Differences

Beyond their handling of infinity, the two theories differ in their execution and scope:

- **Partitioning:** In Riemann's theory, we partition the **domain** of the function, whereas in Lebesgue's theory, we partition the **range**.
- **Generalization:** Riemann's theory is largely restricted to Euclidean space. Conversely, the principles of Lebesgue's theory are applicable to more general spaces, leading to the development of **abstract measure theory**. This abstract approach is highly versatile and intersects with numerous branches of mathematics.

I suggest all students read these two series of notes by **Prof. A. K. Nandakumaran**, Department of Mathematics, IISc Bangalore. These are fantastic articles that provide excellent motivation for studying Measure Theory:

- Meaning of Integration 1
- Meaning of Integration 2

## The Vitali Construction: Limits of Generalizing Volume

In this section, we attempt to generalize the concept of volume (specifically length in  $\mathbb{R}$ ) by searching for a set function  $\mu$  that assigns a "size" to subsets of  $\mathbb{R}$ . We want this function  $\mu$  to satisfy certain natural properties:

1. **Normalization:** The length of the unit interval is one, i.e.,  $\mu([0, 1]) = 1$ .
2. **Countable Additivity:** If  $E_1, E_2, \dots$  are disjoint subsets, then the measure of their union is the sum of their individual measures:  $\mu(\bigcup_{j=1}^{\infty} E_j) = \sum_{j=1}^{\infty} \mu(E_j)$ .
3. **Translation Invariance:** If a set  $E$  is shifted by a distance  $r$  to form a new set  $F = E + r$ , then  $\mu(F) = \mu(E)$ .

The following construction, known as the **Vitali Set**, demonstrates that these three assumptions are mutually inconsistent. It proves that there exist sets for which a consistent "length" cannot be defined.

## Step-by-Step Construction of the Vitali Set

We work within the interval  $[0, 1]$  in  $\mathbb{R}$ .

**Step 1: Define an Equivalence Relation** We define a relation  $\sim$  on  $[0, 1]$ . For any  $a, b \in [0, 1]$ , we say  $a \sim b$  if and only if their difference is a rational number:

$$a \sim b \iff a - b \in \mathbb{Q}$$

This partitions  $[0, 1]$  into disjoint equivalence classes, where each class contains all points that differ from each other by only a rational value.

**Step 2: Form the Vitali Set  $Y$**  Using the **Axiom of Choice**, we construct a set  $Y \subset [0, 1]$  by picking exactly one element from each equivalence class. This set  $Y$  is our "Vitali Set."

**Step 3: Analyze Rational Translations** Let  $Q = \mathbb{Q} \cap [-1, 1]$  be the set of rational numbers between  $-1$  and  $1$ . Since  $\mathbb{Q}$  is countable, we can list these rationals as  $\{r_1, r_2, r_3, \dots\}$ . Consider the translated sets:

$$Y_i = Y + r_i = \{y + r_i : y \in Y\}$$

**Step 4: Verify Disjointness** We claim that if  $r_i \neq r_j$ , then  $Y_i \cap Y_j = \emptyset$ . Suppose there was an element in both. Then  $y_1 + r_i = y_2 + r_j$  for some  $y_1, y_2 \in Y$ . This implies  $y_1 - y_2 = r_j - r_i$ . Since  $r_j$  and  $r_i$  are rational, their difference is rational, meaning  $y_1 \sim y_2$ . However, since  $Y$  contains only one element from each class, we must have  $y_1 = y_2$ , which forces  $r_i = r_j$ . Thus, the translates are disjoint.

**Step 5: Containment and Contradiction** Note that since  $Y \subset [0, 1]$  and each  $r_i \in [-1, 1]$ , every translated set  $Y_i$  is contained within the interval  $[-1, 2]$ . Therefore:

$$\bigcup_i Y_i \subset [-1, 2]$$

By translation invariance, all  $Y_i$  must have the same measure:  $\mu(Y_i) = \mu(Y) = \delta$ . By countable additivity:

$$\mu\left(\bigcup_i Y_i\right) = \sum_{i=1}^{\infty} \mu(Y_i) = \sum_{i=1}^{\infty} \delta$$

- If  $\delta = 0$ , the total measure is 0. But the union of these translates actually covers the interval  $[0, 1]$ , so the measure must be at least 1. (Contradiction:  $0 \geq 1$ )
- If  $\delta > 0$ , the infinite sum  $\sum \delta$  diverges to  $\infty$ . But the union is contained in  $[-1, 2]$ , so the measure cannot exceed 3. (Contradiction:  $\infty \leq 3$ )

Thus, no such function  $\mu$  can exist for all subsets of  $\mathbb{R}$ .

**Step 6: Finite Additivity Failure** Suppose  $\mu(Y) = \delta$ . Since  $Y \subset [0, 1]$ ,  $\delta$  must be some non-negative value.

- If  $\delta = 0$ , then any countable union of its translates would also have measure 0, which cannot cover  $[0, 1]$  (as  $\mu([0, 1]) = 1$ ).
- If  $\delta > 0$ , we can find a natural number  $k$  such that  $k\delta > 2$ .

Now, choose  $k$  distinct rational numbers  $q_1, q_2, \dots, q_k$  from  $\mathbb{Q} \cap [0, 1]$ . Define the translated sets  $Y_i = Y + q_i$ .

1. **Disjointness:** As shown previously, these translates are disjoint because  $Y$  picks only one element per class.
2. **Containment:** Since  $Y \subset [0, 1]$  and  $q_i \in [0, 1]$ , it follows that each  $Y_i \subset [0, 2]$ . Thus:

$$\bigcup_{i=1}^k Y_i \subset [0, 2]$$

3. **The Contradiction:** By monotonicity and normalization,  $\mu(\bigcup_{i=1}^k Y_i) \leq \mu([0, 2]) = 2$ . However, by finite additivity and translation invariance:

$$\mu\left(\bigcup_{i=1}^k Y_i\right) = \sum_{i=1}^k \mu(Y_i) = \sum_{i=1}^k \delta = k\delta$$

Since we chose  $k$  such that  $k\delta > 2$ , we get  $k\delta \leq 2$ , which is a contradiction.

Thus, even finite additivity is inconsistent with translation invariance and normalization.

**Remark.** The set  $Y$  constructed above is an example of a **non-measurable set**. It shows that we cannot assign a meaningful "length" to every possible collection of real numbers if we want to keep the properties of additivity and translation invariance.

**Exercise 1.** Find two disjoint subsets  $A, B \subset [0, 1]$  such that  $\mu(A \cup B) \neq \mu(A) + \mu(B)$ .

*Solution.* Consider the translates  $Y_1, Y_2, \dots, Y_k$  from the construction above. We know that:

$$\mu(Y_1 \cup Y_2 \cup \dots \cup Y_k) \neq \sum_{j=1}^k \mu(Y_j)$$

We can find the specific pair by induction:

1. Check  $Y_1$  and  $Y_2$ . If  $\mu(Y_1 \cup Y_2) \neq \mu(Y_1) + \mu(Y_2)$ , we are done. Set  $A = Y_1, B = Y_2$ .
2. If they are equal, consider the set  $(Y_1 \cup Y_2)$  and  $Y_3$ . If  $\mu((Y_1 \cup Y_2) \cup Y_3) \neq \mu(Y_1 \cup Y_2) + \mu(Y_3)$ , then set  $A = Y_1 \cup Y_2$  and  $B = Y_3$ .
3. Continue this process. We must reach a first natural number  $r \leq k$  such that:

$$\mu\left(\left(\bigcup_{j=1}^{r-1} Y_j\right) \cup Y_r\right) \neq \mu\left(\bigcup_{j=1}^{r-1} Y_j\right) + \mu(Y_r)$$

This stage  $r$  is guaranteed to exist because if additivity held for every step up to  $k$ , the total sum would equal the measure of the union, which we already proved is impossible.

Taking  $A = \bigcup_{j=1}^{r-1} Y_j$  and  $B = Y_r$  provides the required disjoint sets. □

## $\sigma$ -Algebras and Measurable Spaces

**Definition 1.** Let  $X$  be a nonempty set. An **algebra** of sets on  $X$  (also called a **field**) is a nonempty collection  $\mathcal{A}$  of subsets of  $X$  that satisfies the following closure properties:

1. **Closed under finite unions:** If  $E_1, \dots, E_n \in \mathcal{A}$ , then  $\bigcup_{j=1}^n E_j \in \mathcal{A}$ .
2. **Closed under complements:** If  $E \in \mathcal{A}$ , then  $E^c \in \mathcal{A}$ .

**Definition 2.** A  $\sigma$ -**algebra** (or  $\sigma$ -**field**) is an algebra that is also closed under countable unions. That is, if  $\{E_j\}_{j=1}^\infty \subset \mathcal{A}$ , then  $\bigcup_{j=1}^\infty E_j \in \mathcal{A}$ .

**Remark.** By applying De Morgan's laws,  $\bigcap E_j = (\bigcup E_j^c)^c$ , we observe that algebras and  $\sigma$ -algebras are also closed under finite and countable intersections, respectively. Furthermore, any algebra  $\mathcal{A}$  necessarily contains the empty set  $\emptyset$  and the whole set  $X$ , since for any  $E \in \mathcal{A}$ , we have  $\emptyset = E \cap E^c$  and  $X = E \cup E^c$ .

**Proposition.** An algebra  $\mathcal{A}$  is a  $\sigma$ -algebra if and only if it is closed under countable disjoint unions.

*Proof.* Suppose  $\{E_j\}_{j=1}^\infty \subset \mathcal{A}$ . To show the countable union is in  $\mathcal{A}$ , we can define a sequence of disjoint sets  $\{F_k\}$  as follows:

$$F_1 = E_1, \quad F_k = E_k \setminus \left[ \bigcup_{j=1}^{k-1} E_j \right] = E_k \cap \left[ \bigcup_{j=1}^{k-1} E_j \right]^c \quad \text{for } k > 1$$

These  $F_k$  belong to  $\mathcal{A}$  because  $\mathcal{A}$  is an algebra (closed under finite unions and complements). By construction, the  $F_k$  are disjoint and satisfy  $\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty F_j$ . Thus, if  $\mathcal{A}$  is closed under countable disjoint unions, it must also be closed under all countable unions. □

**Remark.** The technique of replacing an arbitrary sequence of sets with a disjoint sequence such that their union remains unchanged is a fundamental tool in measure theory.

**Example 1.** Let  $X$  be any non-empty set. The following collections of subsets are  $\sigma$ -algebras on  $X$ :

1. **The Power Set:**  $\mathcal{P}(X)$ , the collection of all subsets of  $X$ .
2. **The Trivial  $\sigma$ -algebra:**  $\{\emptyset, X\}$ , the smallest possible  $\sigma$ -algebra.
3. **The Countable or Co-countable  $\sigma$ -algebra:** If  $X$  is an uncountable set, then the collection

$$\mathcal{A} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a  $\sigma$ -algebra.

**Remark.** The verification of Example 3 relies on the following logic regarding countable unions:

- **Case 1 (All sets are countable):** If we have a sequence  $\{E_j\}_{j=1}^\infty \subset \mathcal{A}$  where every  $E_j$  is countable, then their union  $\bigcup_{j=1}^\infty E_j$  is a countable union of countable sets, which is itself countable. Thus, the union belongs to  $\mathcal{A}$ .
- **Case 2 (At least one set is co-countable):** If at least one  $E_{j_0}$  is co-countable (meaning  $E_{j_0}^c$  is countable), then the complement of the union  $(\bigcup E_j)^c = \bigcap E_j^c$  is a subset of  $E_{j_0}^c$ . Since  $E_{j_0}^c$  is countable and any subset of a countable set is countable, the complement of the union is countable. Thus, the union is co-countable and belongs to  $\mathcal{A}$ .

**Exercise 2.** Prove that the intersection of any family of  $\sigma$ -algebras on a set  $X$  is again a  $\sigma$ -algebra on  $X$ .

*Solution.* Let  $\{\mathcal{M}_\alpha\}_{\alpha \in I}$  be a family of  $\sigma$ -algebras on  $X$ . Let  $\mathcal{M} = \bigcap_{\alpha \in I} \mathcal{M}_\alpha$ .

1. Since  $X \in \mathcal{M}_\alpha$  for every  $\alpha$ ,  $X \in \mathcal{M}$ .
2. If  $E \in \mathcal{M}$ , then  $E \in \mathcal{M}_\alpha$  for all  $\alpha$ . Since each  $\mathcal{M}_\alpha$  is a  $\sigma$ -algebra,  $E^c \in \mathcal{M}_\alpha$  for all  $\alpha$ , so  $E^c \in \mathcal{M}$ .
3. If  $\{E_j\}_{j=1}^\infty \subset \mathcal{M}$ , then  $\{E_j\}_{j=1}^\infty \subset \mathcal{M}_\alpha$  for all  $\alpha$ . Thus  $\bigcup E_j \in \mathcal{M}_\alpha$  for all  $\alpha$ , implying  $\bigcup E_j \in \mathcal{M}$ .

□

**Definition 3.** Let  $\mathcal{E}$  be any subset of the power set  $\mathcal{P}(X)$ . There exists a unique smallest  $\sigma$ -algebra containing  $\mathcal{E}$ , denoted by  $\mathcal{M}(\mathcal{E})$ . It is defined as the intersection of all  $\sigma$ -algebras on  $X$  that contain  $\mathcal{E}$ :

$$\mathcal{M}(\mathcal{E}) = \bigcap \{\mathcal{M} : \mathcal{M} \text{ is a } \sigma\text{-algebra and } \mathcal{E} \subset \mathcal{M}\}$$

This intersection is well-defined because there is always at least one such  $\sigma$ -algebra, namely  $\mathcal{P}(X)$ .  $\mathcal{M}(\mathcal{E})$  is called the  **$\sigma$ -algebra generated by  $\mathcal{E}$** .

**Lemma.** If  $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$ , then  $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$ .

*Proof.* By definition,  $\mathcal{M}(\mathcal{E})$  is the *smallest*  $\sigma$ -algebra containing the collection of sets  $\mathcal{E}$ . This means that if  $\mathcal{A}$  is any  $\sigma$ -algebra such that  $\mathcal{E} \subset \mathcal{A}$ , then it must follow that  $\mathcal{M}(\mathcal{E}) \subset \mathcal{A}$ .

In this case, we are given that  $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$ . We already know by definition that  $\mathcal{M}(\mathcal{F})$  is a  $\sigma$ -algebra. Therefore,  $\mathcal{M}(\mathcal{F})$  is a  $\sigma$ -algebra that contains  $\mathcal{E}$ . By the minimality of  $\mathcal{M}(\mathcal{E})$  discussed above, we conclude that  $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$ . □

## Further Definitions in $\sigma$ -Algebras

**Definition 4.** Let  $\mathcal{E}$  be a collection of subsets of  $X$ . The  **$\sigma$ -algebra generated by  $\mathcal{E}$** , denoted by  $\mathcal{M}(\mathcal{E})$ , is the unique smallest  $\sigma$ -algebra containing  $\mathcal{E}$ . It is defined as the intersection of all  $\sigma$ -algebras on  $X$  that contain  $\mathcal{E}$ . Since the power set  $\mathcal{P}(X)$  is always one such  $\sigma$ -algebra, this intersection is well-defined.

**Lemma.** If  $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$ , then  $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$ .

*Proof.* By definition,  $\mathcal{M}(\mathcal{F})$  is a  $\sigma$ -algebra. Since it is given that  $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$ , then  $\mathcal{M}(\mathcal{F})$  is one of the  $\sigma$ -algebras included in the intersection that defines  $\mathcal{M}(\mathcal{E})$ . Because  $\mathcal{M}(\mathcal{E})$  is the intersection of all such  $\sigma$ -algebras, it must be a subset of any individual  $\sigma$ -algebra in that collection, specifically  $\mathcal{M}(\mathcal{F})$ .  $\square$

**Definition 5.** Let  $X$  be a topological space (such as a metric space). The  $\sigma$ -algebra generated by the family of all open sets in  $X$  is called the **Borel  $\sigma$ -algebra** on  $X$ , denoted by  $\mathcal{B}_X$ . The members of  $\mathcal{B}_X$  are called **Borel sets**.

**Remark.** Since  $\sigma$ -algebras are closed under complements, the Borel  $\sigma$ -algebra is equivalently generated by the family of all closed sets in  $X$ .  $\mathcal{B}_X$  includes all open sets, closed sets, countable intersections of open sets, and countable unions of closed sets.

**Definition 6.** Within the hierarchy of Borel sets, there is standard terminology for specific levels:

- A  $G_\delta$  set is a countable intersection of open sets.
- An  $F_\sigma$  set is a countable union of closed sets.
- A  $G_{\delta\sigma}$  set is a countable union of  $G_\delta$  sets.
- An  $F_{\sigma\delta}$  set is a countable intersection of  $F_\sigma$  sets.

The symbols  $\delta$  and  $\sigma$  originate from the German words *Durchschnitt* (intersection) and *Summe* (union).

## Generators of the Borel $\sigma$ -algebra on $\mathbb{R}$

**Proposition.** The Borel  $\sigma$ -algebra  $\mathcal{B}_{\mathbb{R}}$  is generated by each of the following collections of intervals:

- a. The open intervals:  $\mathcal{E}_1 = \{(a, b) : a < b\}$
- b. The closed intervals:  $\mathcal{E}_2 = \{[a, b] : a < b\}$
- c. The half-open intervals:  $\mathcal{E}_3 = \{(a, b] : a < b\}$  or  $\mathcal{E}_4 = \{[a, b) : a < b\}$
- d. The open rays:  $\mathcal{E}_5 = \{(a, \infty) : a \in \mathbb{R}\}$  or  $\mathcal{E}_6 = \{(-\infty, a) : a \in \mathbb{R}\}$
- e. The closed rays:  $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$  or  $\mathcal{E}_8 = \{(-\infty, a] : a \in \mathbb{R}\}$

*Proof.* Recall that  $\mathcal{B}_{\mathbb{R}}$  is generated by the collection of all open sets in  $\mathbb{R}$ . We will use Lemma 1 to show that  $\mathcal{M}(\mathcal{E}_j) = \mathcal{B}_{\mathbb{R}}$  for each  $j$ .

**Step 1: Establishing  $\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$**  Since every open interval  $(a, b)$  is an open set, we have  $\mathcal{E}_1 \subset \mathcal{B}_{\mathbb{R}}$ , which implies  $\mathcal{M}(\mathcal{E}_1) \subset \mathcal{B}_{\mathbb{R}}$ . Conversely, every open set in  $\mathbb{R}$  can be written as a countable union of open intervals. Since  $\sigma$ -algebras are closed under countable unions, any open set must belong to  $\mathcal{M}(\mathcal{E}_1)$ . Therefore,  $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_1)$ , and we conclude  $\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$ .

**Step 2: Showing  $\mathcal{M}(\mathcal{E}_j) \subset \mathcal{B}_{\mathbb{R}}$  for all  $j$**  The elements of  $\mathcal{E}_j$  for  $j \neq 1$  are either open or closed sets, which are inherently Borel sets. For  $\mathcal{E}_3$  and  $\mathcal{E}_4$ , notice that:

$$(a, b] = \bigcap_{n=1}^{\infty} \left( a, b + \frac{1}{n} \right) \quad \text{and} \quad [a, b) = \bigcap_{n=1}^{\infty} \left( a - \frac{1}{n}, b \right)$$

Since these are countable intersections of open sets ( $G_\delta$  sets), they are Borel sets. Thus,  $\mathcal{E}_j \subset \mathcal{B}_{\mathbb{R}}$  for all  $j$ , which implies  $\mathcal{M}(\mathcal{E}_j) \subset \mathcal{B}_{\mathbb{R}}$ .

**Step 3: Showing  $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_j)$  for  $j \geq 2$**  To prove this, it suffices to show that any open interval  $(a, b)$  can be constructed using elements from each  $\mathcal{E}_j$ .

- **For  $\mathcal{E}_2$  (Closed intervals):** We can write  $(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}]$ . Since each  $[a + \frac{1}{n}, b - \frac{1}{n}] \in \mathcal{E}_2$ , the union is in  $\mathcal{M}(\mathcal{E}_2)$ .

- **For  $\mathcal{E}_3$  (Half-open  $(a, b]$ ):** We can write  $(a, b) = \bigcup_{n=1}^{\infty} (a, b - \frac{1}{n}]$ . Thus  $\mathcal{E}_1 \subset \mathcal{M}(\mathcal{E}_3)$ .
- **For  $\mathcal{E}_5$  (Open rays  $(a, \infty)$ ):** Note that  $(-\infty, b] = (b, \infty)^c$ , so  $(-\infty, b] \in \mathcal{M}(\mathcal{E}_5)$ . Then  $(a, b] = (a, \infty) \cap (-\infty, b] \in \mathcal{M}(\mathcal{E}_5)$ . Using the result from  $\mathcal{E}_3$ , we get  $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_5)$ .
- **For  $\mathcal{E}_7$  (Closed rays  $[a, \infty)$ ):** Note that  $(a, \infty) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, \infty)$ . Thus  $\mathcal{E}_5 \subset \mathcal{M}(\mathcal{E}_7)$ , which implies  $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_7)$ .

Similar constructions apply to  $\mathcal{E}_4, \mathcal{E}_6$ , and  $\mathcal{E}_8$  by using symmetry and complements. Since  $\mathcal{E}_1 \subset \mathcal{M}(\mathcal{E}_j)$  in all cases, we have  $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_j)$ .  $\square$

## Product $\sigma$ -Algebras

**Definition 7.** Let  $\{X_{\alpha}\}_{\alpha \in A}$  be an indexed collection of nonempty sets, and let  $X = \prod_{\alpha \in A} X_{\alpha}$  denote their Cartesian product. For each  $\alpha \in A$ , let  $\pi_{\alpha} : X \rightarrow X_{\alpha}$  be the coordinate maps (or projection maps). If  $\mathcal{M}_{\alpha}$  is a  $\sigma$ -algebra on  $X_{\alpha}$  for each  $\alpha$ , the **product  $\sigma$ -algebra** on  $X$  is defined as the  $\sigma$ -algebra generated by the following collection:

$$\{\pi_{\alpha}^{-1}(E_{\alpha}) : E_{\alpha} \in \mathcal{M}_{\alpha}, \alpha \in A\}$$

This  $\sigma$ -algebra is denoted by  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ .

**Remark.** In the specific case where the index set is finite, such as  $A = \{1, \dots, n\}$ , we use the notation  $\bigotimes_1^n \mathcal{M}_j$  or  $\mathcal{M}_1 \otimes \dots \otimes \mathcal{M}_n$ . The importance of this definition is primarily established when discussing the measurability of functions on product spaces.

**Proposition.** If the index set  $A$  is countable, then the product  $\sigma$ -algebra  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$  is generated by the collection of measurable rectangles:

$$\mathcal{F} = \left\{ \prod_{\alpha \in A} E_{\alpha} : E_{\alpha} \in \mathcal{M}_{\alpha} \right\}$$

*Proof.* To prove that  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha} = \mathcal{M}(\mathcal{F})$ , we show containment in both directions using Lemma 1.

**Step 1: Showing  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha} \subset \mathcal{M}(\mathcal{F})$**  Let  $\pi_{\alpha}^{-1}(E_{\alpha})$  be a generator of the product  $\sigma$ -algebra, where  $E_{\alpha} \in \mathcal{M}_{\alpha}$ . This inverse image can be written as a special type of product:

$$\pi_{\alpha}^{-1}(E_{\alpha}) = \prod_{\beta \in A} E_{\beta}, \quad \text{where } E_{\beta} = \begin{cases} E_{\alpha} & \text{if } \beta = \alpha \\ X_{\beta} & \text{if } \beta \neq \alpha \end{cases}$$

Since  $X_{\beta}$  is always in  $\mathcal{M}_{\beta}$ , this set is a member of the collection  $\mathcal{F}$ . Thus, every generator of the product  $\sigma$ -algebra is contained in  $\mathcal{F}$ . By Lemma 1, it follows that  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha} \subset \mathcal{M}(\mathcal{F})$ .

**Step 2: Showing  $\mathcal{M}(\mathcal{F}) \subset \bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$**  Now consider an arbitrary element of  $\mathcal{F}$ , which is a product of the form  $\prod_{\alpha \in A} E_{\alpha}$ . We can express this product as an intersection of inverse images:

$$\prod_{\alpha \in A} E_{\alpha} = \bigcap_{\alpha \in A} \pi_{\alpha}^{-1}(E_{\alpha})$$

By the definition of the product  $\sigma$ -algebra, each set  $\pi_{\alpha}^{-1}(E_{\alpha})$  is a generator and thus belongs to  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ . Since the index set  $A$  is countable, this is a **countable intersection** of measurable sets. Since  $\sigma$ -algebras are closed under countable intersections, the resulting product belongs to  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ .

Therefore,  $\mathcal{F} \subset \bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ , which by Lemma 1 implies  $\mathcal{M}(\mathcal{F}) \subset \bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$ . We conclude that the two  $\sigma$ -algebras are identical.  $\square$

**Proposition.** Suppose that for each  $\alpha \in A$ , the  $\sigma$ -algebra  $\mathcal{M}_{\alpha}$  is generated by a collection  $\mathcal{E}_{\alpha}$  (i.e.,  $\mathcal{M}_{\alpha} = \mathcal{M}(\mathcal{E}_{\alpha})$ ). Then the product  $\sigma$ -algebra  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$  is generated by the collection:

$$\mathcal{F}_1 = \{\pi_{\alpha}^{-1}(E_{\alpha}) : E_{\alpha} \in \mathcal{E}_{\alpha}, \alpha \in A\}$$

Furthermore, if the index set  $A$  is countable and  $X_{\alpha} \in \mathcal{E}_{\alpha}$  for all  $\alpha$ , then  $\bigotimes_{\alpha \in A} \mathcal{M}_{\alpha}$  is also generated by the collection of "simple" rectangles:

$$\mathcal{F}_2 = \left\{ \prod_{\alpha \in A} E_{\alpha} : E_{\alpha} \in \mathcal{E}_{\alpha} \right\}$$

*Proof.* We begin by proving the first assertion regarding  $\mathcal{F}_1$ .

**Step 1: Establishing  $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$**  By definition, each  $\mathcal{E}_\alpha \subset \mathcal{M}_\alpha$ . Consequently, every set in  $\mathcal{F}_1$  is of the form  $\pi_\alpha^{-1}(E_\alpha)$  where  $E_\alpha \in \mathcal{M}_\alpha$ . These are precisely the generators of the product  $\sigma$ -algebra. Since  $\mathcal{F}_1 \subset \bigotimes \mathcal{M}_\alpha$ , it follows by Lemma 1 that  $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes \mathcal{M}_\alpha$ .

**Step 2: Establishing  $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$**  For a fixed  $\alpha \in A$ , consider the collection of subsets of  $X_\alpha$  whose inverse images under the coordinate map fall into  $\mathcal{M}(\mathcal{F}_1)$ :

$$\mathcal{G}_\alpha = \{E \subset X_\alpha : \pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)\}$$

We can verify that  $\mathcal{G}_\alpha$  is a  $\sigma$ -algebra on  $X_\alpha$ :

- $X_\alpha \in \mathcal{G}_\alpha$  because  $\pi_\alpha^{-1}(X_\alpha) = X$ , which is in any  $\sigma$ -algebra.
- If  $E \in \mathcal{G}_\alpha$ , then  $\pi_\alpha^{-1}(E^c) = (\pi_\alpha^{-1}(E))^c \in \mathcal{M}(\mathcal{F}_1)$ , so  $E^c \in \mathcal{G}_\alpha$ .
- If  $\{E_j\} \subset \mathcal{G}_\alpha$ , then  $\pi_\alpha^{-1}(\cup E_j) = \cup \pi_\alpha^{-1}(E_j) \in \mathcal{M}(\mathcal{F}_1)$ , so  $\cup E_j \in \mathcal{G}_\alpha$ .

By the construction of  $\mathcal{F}_1$ , we know that  $\mathcal{E}_\alpha \subset \mathcal{G}_\alpha$ . Since  $\mathcal{M}_\alpha$  is the *smallest*  $\sigma$ -algebra containing  $\mathcal{E}_\alpha$ , we must have  $\mathcal{M}_\alpha \subset \mathcal{G}_\alpha$ . This implies that for *any*  $E \in \mathcal{M}_\alpha$ , the set  $\pi_\alpha^{-1}(E)$  belongs to  $\mathcal{M}(\mathcal{F}_1)$ . Since these inverse images generate the product  $\sigma$ -algebra, we conclude  $\bigotimes \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$ .

**Step 3: Proving the second assertion for  $\mathcal{F}_2$**  If  $A$  is countable and  $X_\alpha \in \mathcal{E}_\alpha$ , we can use the same logic as the previous Proposition. Every element  $\pi_\alpha^{-1}(E_\alpha)$  in  $\mathcal{F}_1$  can be written as a product  $\prod E_\beta$  where  $E_\beta = X_\beta$  for  $\beta \neq \alpha$ . Since  $X_\beta \in \mathcal{E}_\beta$ , this product is in  $\mathcal{F}_2$ . Thus  $\mathcal{F}_1 \subset \mathcal{F}_2$ , and conversely, every element of  $\mathcal{F}_2$  is a countable intersection of elements from  $\mathcal{F}_1$ :

$$\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha)$$

The result then follows directly from the closure of  $\sigma$ -algebras under countable intersections. □

## Borel Sets on Product Spaces

### Preliminary: Separability and Product Metrics

Before relating product  $\sigma$ -algebras to Borel  $\sigma$ -algebras, we recall some essential concepts from the theory of metric spaces.

**Definition 8.** A metric space  $(X, \rho)$  is called **separable** if it contains a countable dense subset. That is, there exists a set  $C = \{c_1, c_2, \dots\}$  such that for every  $x \in X$  and every  $\epsilon > 0$ , the open ball  $B(x, \epsilon)$  contains at least one point of  $C$ .

**Example 2.** The space  $\mathbb{R}^n$  with the usual Euclidean metric is separable. The set  $\mathbb{Q}^n = \{(q_1, \dots, q_n) : q_i \in \mathbb{Q}\}$  is a countable dense subset of  $\mathbb{R}^n$ .

**Remark.** An important property of separable metric spaces is that they possess a **countable base**. Let  $C$  be a countable dense subset of  $X$ . Let  $\mathcal{E}$  be the collection of all open balls with centers in  $C$  and rational radii:

$$\mathcal{E} = \{B(c, r) : c \in C, r \in \mathbb{Q}^+\}$$

Since  $C$  and  $\mathbb{Q}^+$  are countable,  $\mathcal{E}$  is a countable collection. Every open set  $V \subset X$  can be written as a countable union of members of  $\mathcal{E}$ . *Proof idea:* For any  $x \in V$ , there is a ball  $B(x, \epsilon) \subseteq V$ . By density, we can find  $c_i \in C$  very close to  $x$  and a rational radius  $r$  such that  $x \in B(c_i, r) \subseteq B(x, \epsilon) \subseteq V$ . Thus,  $V$  is the union of all such balls from  $\mathcal{E}$  contained within it.

**Definition 9.** Let  $(X_1, \rho_1), \dots, (X_n, \rho_n)$  be metric spaces. The **product metric**  $\rho$  on  $X = \prod_{j=1}^n X_j$  is typically defined as:

$$\rho((x_1, \dots, x_n), (y_1, \dots, y_n)) = \max_{1 \leq j \leq n} \{\rho_j(x_j, y_j)\}$$

Under this metric, an open ball in  $X$  is simply the product of open balls in the component spaces:

$$B_X(x, r) = B_{X_1}(x_1, r) \times B_{X_2}(x_2, r) \times \dots \times B_{X_n}(x_n, r)$$

Furthermore, if each  $(X_j, \rho_j)$  is separable with countable dense subset  $C_j$ , then  $X$  is separable with countable dense subset  $C = \prod_{j=1}^n C_j$ .

### The Relationship between $\mathcal{B}_X$ and $\bigotimes \mathcal{B}_{X_j}$

**Proposition.** Let  $X_1, \dots, X_n$  be metric spaces and let  $X = \prod_1^n X_j$  be equipped with the product metric. Then  $\bigotimes_1^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$ . If each  $X_j$  is separable, then  $\bigotimes_1^n \mathcal{B}_{X_j} = \mathcal{B}_X$ .

*Proof.* We approach the proof in steps to distinguish between the general case and the case requiring separability.

**Step 1: Understanding the Two  $\sigma$ -Algebras** We are comparing two distinct ways of constructing a  $\sigma$ -algebra on the product space  $X$ :

- **The Borel  $\sigma$ -algebra  $\mathcal{B}_X$ :** This is generated by the collection of all open sets in the product metric space  $X$ . This topology is equivalent to the product topology.
- **The Product  $\sigma$ -algebra  $\bigotimes \mathcal{B}_{X_j}$ :** This is generated by "measurable cylinders" of the form  $\pi_j^{-1}(E_j)$ , where  $E_j \in \mathcal{B}_{X_j}$ .

**Step 2: Proving the General Inclusion  $\bigotimes_1^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$**  By our previous results, the product  $\sigma$ -algebra  $\bigotimes \mathcal{B}_{X_j}$  is generated by the collection:

$$\mathcal{F} = \{\pi_j^{-1}(U_j) : U_j \text{ is open in } X_j, 1 \leq j \leq n\}$$

For each  $j$ , the projection map  $\pi_j$  is continuous. Therefore, if  $U_j$  is an open set in  $X_j$ , its inverse image  $\pi_j^{-1}(U_j)$  is an open set in the product space  $X$ . Since  $\mathcal{B}_X$  is the  $\sigma$ -algebra generated by *all* open sets in  $X$ , it must contain this sub-collection of open sets. By the minimality of generated  $\sigma$ -algebras (Lemma 1), we conclude that  $\bigotimes \mathcal{B}_{X_j} \subset \mathcal{B}_X$ .

**Step 3: Leveraging Separability in  $X_j$**  Suppose each  $X_j$  is separable. Let  $C_j$  be a countable dense subset of  $X_j$ . Let  $\mathcal{E}_j$  be the countable collection of open balls in  $X_j$  with centers in  $C_j$  and rational radii. As noted in our preliminary remarks, every open set in  $X_j$  is a countable union of balls from  $\mathcal{E}_j$ . This implies that the Borel  $\sigma$ -algebra  $\mathcal{B}_{X_j}$  is generated by the countable collection  $\mathcal{E}_j$ .

**Step 4: Completion of the Equality via Separability** Now consider the product space  $X$ . Since  $X$  is a product of finitely many separable spaces, it is also separable with a countable dense subset  $C = \prod C_j$ .

1. **Generation of  $\mathcal{B}_X$ :** Because  $X$  is separable,  $\mathcal{B}_X$  is generated by the collection of all "product balls":

$$\mathcal{F}_2 = \left\{ \prod_{j=1}^n E_j : E_j \in \mathcal{E}_j \right\}$$

This is because any open set in  $X$  is a countable union of such product balls.

2. **Generation of the Product  $\sigma$ -algebra:** From our previous Proposition regarding generators, since  $X_j \in \mathcal{E}_j$  (by taking a sufficiently large rational radius) and the index set is finite (and thus countable), the product  $\sigma$ -algebra  $\bigotimes \mathcal{B}_{X_j}$  is also generated by the same collection of "simple" rectangles  $\mathcal{F}_2$ .

Since both  $\mathcal{B}_X$  and  $\bigotimes \mathcal{B}_{X_j}$  are generated by the exact same collection of sets  $\mathcal{F}_2$ , they must be identical.  $\square$

**Remark.** The assumption of separability is crucial. In non-separable spaces, the Borel  $\sigma$ -algebra  $\mathcal{B}_X$  can be strictly larger than the product  $\sigma$ -algebra because there may be open sets in  $X$  that cannot be represented as countable unions of measurable rectangles.

**Corollary 1.**  $\mathcal{B}_{\mathbb{R}^n} = \bigotimes_1^n \mathcal{B}_{\mathbb{R}}$ .

*Proof.* This follows directly from the previous Proposition. Since  $\mathbb{R}$  is a separable metric space (containing the countable dense subset  $\mathbb{Q}$ ), its  $n$ -fold Cartesian product  $\mathbb{R}^n$  satisfies the condition for the Borel  $\sigma$ -algebra to coincide with the product  $\sigma$ -algebra.  $\square$



## Elementary Family

**Definition 10.** A collection  $\mathcal{E}$  of subsets of  $X$  is called an **elementary family** if it satisfies the following three properties:

1.  $\emptyset \in \mathcal{E}$ .
2. If  $E, F \in \mathcal{E}$ , then  $E \cap F \in \mathcal{E}$  (closed under finite intersections).
3. If  $E \in \mathcal{E}$ , then its complement  $E^c$  is a finite disjoint union of members of  $\mathcal{E}$ . That is,  $E^c = \bigsqcup_{j=1}^n E_j$  for some  $E_1, \dots, E_n \in \mathcal{E}$ .

**Proposition.** If  $\mathcal{E}$  is an elementary family, the collection  $\mathcal{A}$  of finite disjoint unions of members of  $\mathcal{E}$  is an algebra.

*Proof.* To prove that  $\mathcal{A}$  is an algebra, we must show that it is closed under finite unions and complements. Recall that an element of  $\mathcal{A}$  is of the form  $\bigsqcup_{i=1}^n E_i$  where each  $E_i \in \mathcal{E}$ .

**Step 1: Closed under Finite Unions** First, we observe that if we can show the union of two elements of  $\mathcal{E}$  belongs to  $\mathcal{A}$ , the general case for elements of  $\mathcal{A}$  follows.

Let  $A, B \in \mathcal{E}$ . We want to show  $A \cup B \in \mathcal{A}$ . Since  $\mathcal{E}$  is an elementary family, the complement  $B^c$  can be written as a finite disjoint union of members of  $\mathcal{E}$ , say  $B^c = \bigsqcup_{j=1}^J C_j$  where  $C_j \in \mathcal{E}$ .

- We can write the set difference as  $A \setminus B = A \cap B^c = A \cap (\bigsqcup_{j=1}^J C_j) = \bigsqcup_{j=1}^J (A \cap C_j)$ .
- Since  $\mathcal{E}$  is closed under finite intersections, each  $(A \cap C_j) \in \mathcal{E}$ . Because the  $C_j$  are disjoint, these intersections are also disjoint. Thus,  $A \setminus B \in \mathcal{A}$ .
- Now, we can write the union as  $A \cup B = (A \setminus B) \sqcup B$ . Since both  $(A \setminus B)$  and  $B$  are finite disjoint unions of elements of  $\mathcal{E}$ , their union is also a finite disjoint union of elements of  $\mathcal{E}$ . Hence,  $A \cup B \in \mathcal{A}$ .

By induction, it follows that if  $A_1, \dots, A_n \in \mathcal{E}$ , then  $\bigcup_1^n A_j \in \mathcal{A}$ . Specifically, for the inductive step, we assume the union of  $n - 1$  sets is in  $\mathcal{A}$  and write:

$$\bigcup_{j=1}^n A_j = A_n \cup \bigcup_{j=1}^{n-1} (A_j \setminus A_n)$$

This expresses the union as a disjoint union of elements in  $\mathcal{A}$ , confirming that  $\mathcal{A}$  is closed under finite unions.

**Step 2: Closed under Complements** Let  $A \in \mathcal{A}$ . By definition,  $A = \bigcup_{m=1}^n A_m$  where  $A_m \in \mathcal{E}$  and the  $A_m$  are disjoint. We need to show  $A^c \in \mathcal{A}$ . Using De Morgan's Laws:

$$A^c = \left( \bigcup_{m=1}^n A_m \right)^c = \bigcap_{m=1}^n A_m^c$$

Since  $\mathcal{E}$  is an elementary family, each  $A_m^c$  is a finite disjoint union of members of  $\mathcal{E}$ . Let  $A_m^c = \bigsqcup_{j=1}^{J_m} B_m^j$ , where  $B_m^j \in \mathcal{E}$ . Substituting this into the intersection gives:

$$A^c = \bigcap_{m=1}^n \left( \bigcup_{j=1}^{J_m} B_m^j \right)$$

By the distributive law for sets, this intersection of unions can be rewritten as a union of intersections:

$$A^c = \bigcup \{ B_1^{j_1} \cap B_2^{j_2} \cap \dots \cap B_n^{j_n} : 1 \leq j_m \leq J_m \text{ for each } m \}$$

Each term  $B_1^{j_1} \cap \dots \cap B_n^{j_n}$  is an intersection of elements of  $\mathcal{E}$ , so it belongs to  $\mathcal{E}$ . Furthermore, these terms are disjoint because if we change any index  $j_m$ , we are intersecting with a disjoint set from the  $m$ -th family. Thus,  $A^c$  is a finite disjoint union of members of  $\mathcal{E}$ , meaning  $A^c \in \mathcal{A}$ .  $\square$

**Example 3.** Let  $\mathcal{E}$  be the collection of all half-open intervals of the form  $(a, b]$  in  $\mathbb{R}$ , including  $(a, a] = \emptyset$  and infinite intervals like  $(a, \infty)$  and  $(-\infty, b]$ . This collection forms an **elementary family** because:

1.  $\emptyset \in \mathcal{E}$  (by setting  $a = b$ ).
2. The intersection of any two such intervals  $(a, b] \cap (c, d] = (\max(a, c), \min(b, d)]$  is either another half-open interval or empty.
3. The complement of any interval  $(a, b]^c = (-\infty, a] \cup (b, \infty)$  is a finite disjoint union of two elements from  $\mathcal{E}$ .

## Measures and Measure Spaces

**Definition 11.** Let  $X$  be a set equipped with a  $\sigma$ -algebra  $\mathcal{M}$ . A **measure** on  $\mathcal{M}$  (or on  $(X, \mathcal{M})$ , or simply on  $X$  if  $\mathcal{M}$  is understood) is a function  $\mu : \mathcal{M} \rightarrow [0, \infty]$  that satisfies the following two properties:

- i. **Null Empty Set:**  $\mu(\emptyset) = 0$ .
- ii. **Countable Additivity:** If  $\{E_j\}_{j=1}^{\infty}$  is a sequence of disjoint sets in  $\mathcal{M}$ , then:

$$\mu \left( \bigcup_{j=1}^{\infty} E_j \right) = \sum_{j=1}^{\infty} \mu(E_j)$$

**Remark.** Property (ii) implies **finite additivity**. If  $E_1, \dots, E_n$  are disjoint sets in  $\mathcal{M}$ , then:

$$\mu \left( \bigcup_{j=1}^n E_j \right) = \sum_{j=1}^n \mu(E_j)$$

This is easily seen by taking  $E_j = \emptyset$  for all  $j > n$ . A function  $\mu$  that satisfies property (i) and finite additivity, but not necessarily countable additivity, is called a **finitely additive measure**.

**Definition 12.** We use the following standard terminology for the structures involving measures:

- The pair  $(X, \mathcal{M})$  is called a **measurable space**.
- The elements of the  $\sigma$ -algebra  $\mathcal{M}$  are called **measurable sets**.
- The triple  $(X, \mathcal{M}, \mu)$  is called a **measure space**.

## Classification of Measures by Size

Let  $(X, \mathcal{M}, \mu)$  be a measure space. We classify the measure  $\mu$  (and sets within  $\mathcal{M}$ ) based on their "size" as follows:

**Definition 13.** A measure  $\mu$  is called **finite** if  $\mu(X) < \infty$ . Note: Since  $\mu(X) = \mu(E) + \mu(E^c)$  for any  $E \in \mathcal{M}$ , the finiteness of  $\mu(X)$  implies that  $\mu(E) < \infty$  for all measurable sets  $E$ .

**Definition 14.** A measure  $\mu$  is called  **$\sigma$ -finite** if  $X$  can be written as a countable union of measurable sets with finite measure. That is,  $X = \bigcup_{j=1}^{\infty} E_j$  where  $E_j \in \mathcal{M}$  and  $\mu(E_j) < \infty$  for all  $j$ .

**Definition 15.** A set  $E \in \mathcal{M}$  is said to be  **$\sigma$ -finite for  $\mu$**  if  $E = \bigcup_{j=1}^{\infty} E_j$  where  $E_j \in \mathcal{M}$  and  $\mu(E_j) < \infty$  for all  $j$ .

**Definition 16.** A measure  $\mu$  is called **semifinite** if for every  $E \in \mathcal{M}$  with  $\mu(E) = \infty$ , there exists a measurable subset  $F \subset E$  such that  $0 < \mu(F) < \infty$ .

**Theorem 2.** Let  $(X, \mathcal{M}, \mu)$  be a measure space.

- a. (**Monotonicity**) If  $E, F \in \mathcal{M}$  and  $E \subset F$ , then  $\mu(E) \leq \mu(F)$ .
- b. (**Subadditivity**) If  $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$ , then  $\mu(\bigcup_{j=1}^{\infty} E_j) \leq \sum_{j=1}^{\infty} \mu(E_j)$ .
- c. (**Continuity from below**) If  $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$  and  $E_1 \subset E_2 \subset \dots$ , then:

$$\mu \left( \bigcup_{j=1}^{\infty} E_j \right) = \lim_{j \rightarrow \infty} \mu(E_j)$$

d. (**Continuity from above**) If  $\{E_j\}_{j=1}^\infty \subset \mathcal{M}$ ,  $E_1 \supset E_2 \supset \dots$ , and  $\mu(E_1) < \infty$ , then:

$$\mu\left(\bigcap_{j=1}^\infty E_j\right) = \lim_{j \rightarrow \infty} \mu(E_j)$$

*Proof.* **(a) Proof of Monotonicity:** Suppose  $E, F \in \mathcal{M}$  with  $E \subset F$ . We can decompose  $F$  into two disjoint sets:  $E$  and the set difference  $F \setminus E$ . Thus,  $F = E \cup (F \setminus E)$ . By the finite additivity of measures:

$$\mu(F) = \mu(E) + \mu(F \setminus E)$$

Since the range of  $\mu$  is  $[0, \infty]$ , we know  $\mu(F \setminus E) \geq 0$ . Therefore,  $\mu(F) \geq \mu(E)$ .

**(b) Proof of Subadditivity:** To use the additivity property of measures on a sequence that might not be disjoint, we must first "disjointify" it. Let  $\{E_j\}_{j=1}^\infty$  be any sequence in  $\mathcal{M}$ . Define:

$$F_1 = E_1, \quad F_k = E_k \setminus \left(\bigcup_{j=1}^{k-1} E_j\right) \text{ for } k > 1$$

These  $F_k$  are disjoint by construction, and their union is the same as the union of the  $E_j$ 's (i.e.,  $\bigcup_{j=1}^\infty F_j = \bigcup_{j=1}^\infty E_j$ ). Furthermore, for each  $j$ ,  $F_j \subset E_j$ . By countable additivity and the monotonicity proved in (a):

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \mu\left(\bigcup_{j=1}^\infty F_j\right) = \sum_{j=1}^\infty \mu(F_j) \leq \sum_{j=1}^\infty \mu(E_j)$$

**(c) Proof of Continuity from Below:** Let  $E_1 \subset E_2 \subset \dots$  be an increasing sequence of sets. We define  $E_0 = \emptyset$ . We can represent the union as a disjoint union of "rings":

$$\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty (E_j \setminus E_{j-1})$$

Applying countable additivity:

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty \mu(E_j \setminus E_{j-1})$$

By the definition of an infinite series, this is the limit of partial sums:

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(E_j \setminus E_{j-1})$$

Since the sets are nested, the finite sum  $\sum_{j=1}^n \mu(E_j \setminus E_{j-1})$  is a telescoping sum that simplifies to  $\mu(E_n)$ . Thus:

$$\mu\left(\bigcup_{j=1}^\infty E_j\right) = \lim_{n \rightarrow \infty} \mu(E_n)$$

**(d) Proof of Continuity from Above:** Let  $E_1 \supset E_2 \supset \dots$  with  $\mu(E_1) < \infty$ . We transform this into an increasing sequence to apply part (c). Let  $F_j = E_1 \setminus E_j$ . Since the  $E_j$  are decreasing, the  $F_j$  are increasing ( $F_1 \subset F_2 \subset \dots$ ). Note that  $\bigcup F_j = E_1 \setminus (\bigcap E_j)$ . By part (c):

$$\mu(E_1 \setminus \bigcap E_j) = \lim_{j \rightarrow \infty} \mu(F_j)$$

Using the property  $\mu(A \setminus B) = \mu(A) - \mu(B)$  (valid here since  $\mu(E_1) < \infty$ ):

$$\mu(E_1) - \mu\left(\bigcap_{j=1}^\infty E_j\right) = \lim_{j \rightarrow \infty} [\mu(E_1) - \mu(E_j)]$$

Because  $\mu(E_1)$  is finite, we can subtract it from both sides of the equation:

$$-\mu\left(\bigcap_{j=1}^\infty E_j\right) = -\lim_{j \rightarrow \infty} \mu(E_j)$$

Multiplying by  $-1$  gives the desired result:  $\mu(\bigcap_{j=1}^\infty E_j) = \lim_{j \rightarrow \infty} \mu(E_j)$ . □

**Remark.** The condition  $\mu(E_1) < \infty$  in part (d) of Theorem 1.8 is a sufficient but not strictly necessary condition for the first term of the sequence; it could be replaced by  $\mu(E_n) < \infty$  for any  $n > 1$ . This is because the first  $n - 1$  sets in the sequence do not affect the ultimate intersection  $\bigcap_{j=1}^{\infty} E_j$ .

However, **some** finiteness assumption is absolutely necessary. It is possible for  $\mu(E_j) = \infty$  for all  $j$ , yet  $\mu(\bigcap_{j=1}^{\infty} E_j) < \infty$ . In such cases, the limit of the measures may not equal the measure of the intersection.

**Example 4** (Failure of Continuity from Above without Finiteness). Let  $\mu$  be the **counting measure** on the measurable space  $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ . Define a decreasing sequence of sets as follows:

$$E_j = \{n \in \mathbb{N} : n \geq j\} = \{j, j+1, j+2, \dots\}$$

- For every  $j \in \mathbb{N}$ , the set  $E_j$  contains infinitely many elements, so  $\mu(E_j) = \infty$ . This implies:

$$\lim_{j \rightarrow \infty} \mu(E_j) = \infty$$

- However, the intersection of all these sets is empty:  $\bigcap_{j=1}^{\infty} E_j = \emptyset$ .
- Therefore,  $\mu(\bigcap_{j=1}^{\infty} E_j) = \mu(\emptyset) = 0$ .

In this case,  $0 \neq \infty$ , demonstrating that the property fails when the finiteness condition is violated.

**Remark.** Every  $\sigma$ -finite measure is semifinite (Exercise 1.13), but not conversely. Most measures that arise in practice are  $\sigma$ -finite, which is fortunate since non- $\sigma$ -finite measures tend to exhibit pathological behavior.

**Exercise 3** (Exercise 1.13). Prove that every  $\sigma$ -finite measure is semifinite.

*Proof.* Let  $(X, \mathcal{M}, \mu)$  be a  $\sigma$ -finite measure space. By definition,  $X = \bigcup_{n=1}^{\infty} E_n$  where  $E_n \in \mathcal{M}$  and  $\mu(E_n) < \infty$  for all  $n$ . To show  $\mu$  is semifinite, let  $E \in \mathcal{M}$  be such that  $\mu(E) = \infty$ . We must find a measurable subset  $F \subset E$  with  $0 < \mu(F) < \infty$ .

- We can decompose  $E$  using the sequence  $\{E_n\}$ :

$$E = E \cap X = E \cap \left( \bigcup_{n=1}^{\infty} E_n \right) = \bigcup_{n=1}^{\infty} (E \cap E_n)$$

- By subadditivity,  $\mu(E) \leq \sum_{n=1}^{\infty} \mu(E \cap E_n)$ . Since  $\mu(E) = \infty$ , the sum  $\sum \mu(E \cap E_n)$  must also be infinite.
- This implies that there must exist at least one index  $k$  for which  $\mu(E \cap E_k) > 0$ .
- Furthermore, since  $E \cap E_k \subseteq E_k$ , by monotonicity we have  $\mu(E \cap E_k) \leq \mu(E_k) < \infty$ .
- Thus,  $F = E \cap E_k$  is a measurable subset of  $E$  that satisfies  $0 < \mu(F) < \infty$ .

Therefore,  $\mu$  is semifinite. □

## Measures Defined by Functions

A common way to construct measures on a set  $X$  is by using a non-negative function. Let  $X$  be any nonempty set, and let  $\mathcal{M} = \mathcal{P}(X)$  be its power set. Let  $f : X \rightarrow [0, \infty]$  be any function.

**Definition 17.** The function  $f$  determines a measure  $\mu$  on  $\mathcal{M}$  defined by the formula:

$$\mu(E) = \sum_{x \in E} f(x)$$

For the definition of such possibly uncountable sums, one typically defines  $\sum_{x \in E} f(x) = \sup\{\sum_{x \in F} f(x) : F \subseteq E, F \text{ is finite}\}$ .

**Remark.** The properties of the measure  $\mu$  are directly related to the behavior of the function  $f$ :

- $\mu$  is **semifinite** if and only if  $f(x) < \infty$  for every  $x \in X$ .
- $\mu$  is  **$\sigma$ -finite** if and only if  $\mu$  is semifinite and the set  $\{x : f(x) > 0\}$  is countable.

## Special Cases of Significance

Two specific instances of this construction are used throughout measure theory and probability:

1. **Counting Measure:** If  $f(x) = 1$  for all  $x \in X$ , then  $\mu$  is called the **counting measure**.

$$\mu(E) = \begin{cases} |E| & \text{if } E \text{ is finite} \\ \infty & \text{if } E \text{ is infinite} \end{cases}$$

Note that the counting measure is  $\sigma$ -finite if and only if  $X$  is a countable set.

2. **Dirac Measure (Point Mass):** If, for some fixed  $x_0 \in X$ , the function  $f$  is defined by  $f(x_0) = 1$  and  $f(x) = 0$  for  $x \neq x_0$ , then  $\mu$  is called the **point mass** or **Dirac measure** at  $x_0$ , often denoted by  $\delta_{x_0}$ .

$$\delta_{x_0}(E) = \begin{cases} 1 & \text{if } x_0 \in E \\ 0 & \text{if } x_0 \notin E \end{cases}$$

The Dirac measure is always a finite measure (since  $\mu(X) = 1$ ), and thus it is always  $\sigma$ -finite and semifinite.

*Note: These same names are also applied to the restrictions of these measures to smaller  $\sigma$ -algebras on  $X$ .*

**Semifinite measure which is not  $\sigma$ -finite :** To show that a semifinite measure is not necessarily  $\sigma$ -finite, consider the **counting measure**  $\mu$  on an uncountable set  $X$  (for example,  $X = \mathbb{R}$ ) with the  $\sigma$ -algebra  $\mathcal{P}(X)$ .

1.  **$\mu$  is semifinite:** Let  $E \subset X$  with  $\mu(E) = \infty$ . This means  $E$  is an infinite set. Since  $X$  consists of points, we can pick any single point  $x \in E$ . The set  $F = \{x\}$  is a measurable subset of  $E$ , and  $\mu(F) = 1$ . Since  $0 < 1 < \infty$ , the measure is semifinite.
2.  **$\mu$  is not  $\sigma$ -finite:** Suppose  $X = \bigcup_{n=1}^{\infty} E_n$  where  $\mu(E_n) < \infty$  for all  $n$ . Under the counting measure,  $\mu(E_n) < \infty$  means that each  $E_n$  must be a finite set. However, a countable union of finite sets is at most countable. Since we assumed  $X$  is uncountable, no such sequence  $\{E_n\}$  can exist.

Thus,  $\mu$  is semifinite but not  $\sigma$ -finite.

## More Examples

**Example 5** (The Countable/Co-countable Measure). Let  $X$  be an uncountable set, and let  $\mathcal{M}$  be the  $\sigma$ -algebra of countable or co-countable sets. Define  $\mu$  on  $\mathcal{M}$  by:

$$\mu(E) = \begin{cases} 0 & \text{if } E \text{ is countable} \\ 1 & \text{if } E \text{ is co-countable} \end{cases}$$

*This is a valid measure because in any sequence of disjoint sets  $\{E_j\} \subset \mathcal{M}$ , at most one  $E_j$  can be co-countable. This ensures that countable additivity holds.*

**Example 6** (A Finitely Additive Function that is Not a Measure). Let  $X$  be an infinite set and  $\mathcal{M} = \mathcal{P}(X)$ . Define  $\mu(E) = 0$  if  $E$  is finite and  $\mu(E) = \infty$  if  $E$  is infinite.

- **Finite Additivity:** If  $A, B$  are disjoint and finite,  $A \cup B$  is finite ( $0 + 0 = 0$ ). If one is infinite,  $A \cup B$  is infinite ( $\infty + 0 = \infty$ ).
- **Failure of Countable Additivity:** Let  $X = \mathbb{N}$  and  $E_j = \{j\}$ . Then  $\sum \mu(E_j) = \sum 0 = 0$ , but  $\mu(\bigcup E_j) = \mu(\mathbb{N}) = \infty$ .

Thus,  $\mu$  is finitely additive but is **not** a measure.

**Definition 18** (Null Set). If  $(X, \mathcal{M}, \mu)$  is a measure space, a set  $E \in \mathcal{M}$  such that  $\mu(E) = 0$  is called a **null set**. By subadditivity, any countable union of null sets is a null set.

**Definition 19** (Almost Everywhere). If a statement about points  $x \in X$  is true except for  $x$  in some null set, we say that it is true **almost everywhere** (abbreviated **a.e.**), or for **almost every**  $x$ . If more precision is needed, we shall speak of a  **$\mu$ -null set**, or  **$\mu$ -almost everywhere**.

**Definition 20** ( $\sigma$ -finite Set). Let  $(X, \mathcal{M}, \mu)$  be a measure space. A set  $E \in \mathcal{M}$  is said to be  **$\sigma$ -finite** for  $\mu$  if  $E = \bigcup_1^\infty E_j$  where  $E_j \in \mathcal{M}$  and  $\mu(E_j) < \infty$  for all  $j$ .

**Definition 21** (Semifinite Measure). A measure  $\mu$  is called **semifinite** if for each  $E \in \mathcal{M}$  with  $\mu(E) = \infty$ , there exists some  $F \in \mathcal{M}$  with  $F \subset E$  and  $0 < \mu(F) < \infty$ .

**Definition 22** (Finitely Additive Measure). A function  $\mu$  that satisfies  $\mu(\emptyset) = 0$  and finite additivity ( $\mu(\bigcup_1^n E_j) = \sum_1^n \mu(E_j)$  for disjoint sets), but not necessarily countable additivity, is called a **finitely additive measure**.

## Complete Measures

In a general measure space, if  $E$  is a **null set** ( $\mu(E) = 0$ ) and  $F \subset E$ , it follows from monotonicity that  $\mu(F) = 0$  provided that  $F \in \mathcal{M}$ . However, it is not always true that every subset of a null set belongs to the  $\sigma$ -algebra  $\mathcal{M}$ .

**Definition 23.** A measure whose domain includes all subsets of null sets is called **complete**.

Completeness can often simplify technical arguments and can always be achieved by enlarging the domain of the measure  $\mu$ .

**Theorem 3** (Completion of a Measure). Suppose that  $(X, \mathcal{M}, \mu)$  is a measure space. Let  $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$  be the collection of null sets, and define:

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$$

Then  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra, and there is a unique extension  $\bar{\mu}$  of  $\mu$  to a complete measure on  $\overline{\mathcal{M}}$ .

*Proof.* Let  $(X, \mathcal{M}, \mu)$  be a measure space and  $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$ . We define:

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$$

### Step 1: Closed under Countable Unions

Let  $\{A_j\}_{j=1}^\infty$  be a sequence in  $\overline{\mathcal{M}}$ . By definition,  $A_j = E_j \cup F_j$  where  $E_j \in \mathcal{M}$  and  $F_j \subset N_j \in \mathcal{N}$ . We look at the union  $A = \bigcup A_j$ :

- Let  $E = \bigcup E_j$ . Since  $\mathcal{M}$  is a  $\sigma$ -algebra,  $E \in \mathcal{M}$ .
- Let  $F = \bigcup F_j$  and  $N = \bigcup N_j$ . Since  $\mathcal{M}$  is a  $\sigma$ -algebra,  $N \in \mathcal{M}$ .
- By subadditivity,  $\mu(N) \leq \sum \mu(N_j) = 0$ , so  $N \in \mathcal{N}$ .
- Since  $F_j \subset N_j$ , it follows that  $F \subset N$ .

Thus  $A = E \cup F \in \overline{\mathcal{M}}$ .

### Step 2: Closed under Complements

Let  $A = E \cup F \in \overline{\mathcal{M}}$  with  $F \subset N \in \mathcal{N}$ . We can assume  $E \cap N = \emptyset$  by replacing  $F$  and  $N$  with  $F \setminus E$  and  $N \setminus E$  if necessary. Observe that  $E \cup F = (E \cup N) \cap (N^c \cup F)$ . Taking the complement:

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$$

- $(E \cup N)^c \in \mathcal{M}$  because  $E, N \in \mathcal{M}$ .
- $N \setminus F \subset N$ , and since  $N$  is a null set,  $N \setminus F$  is a subset of a null set.

Thus  $A^c \in \overline{\mathcal{M}}$ , and  $\overline{\mathcal{M}}$  is a  $\sigma$ -algebra.

### Step 3: Well-definedness of $\bar{\mu}$

Define  $\bar{\mu}(E \cup F) = \mu(E)$ . Suppose  $E_1 \cup F_1 = E_2 \cup F_2$  with  $F_j \subset N_j \in \mathcal{N}$ . Then:

$$E_1 \subset E_1 \cup F_1 = E_2 \cup F_2 \subset E_2 \cup N_2$$

By monotonicity:  $\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$ . By symmetry,  $\mu(E_2) \leq \mu(E_1)$ . Hence  $\mu(E_1) = \mu(E_2)$ , so  $\bar{\mu}$  is well-defined.

**Step 4: Completeness of  $\bar{\mu}$** 

Suppose  $\bar{\mu}(A) = 0$  for some  $A \in \bar{\mathcal{M}}$ , and let  $B \subset A$ .  $A = E \cup F$  where  $\mu(E) = 0$ . Since  $F \subset N \in \mathcal{N}$ ,  $A \subset E \cup N$ . Let  $N' = E \cup N \in \mathcal{N}$ . Since  $B \subset A \subset N'$ ,  $B$  is a subset of a null set. We can write  $B = \emptyset \cup B$ , so  $B \in \bar{\mathcal{M}}$ . Thus  $\bar{\mu}$  is complete.

**Step 5: Uniqueness**

If  $\nu$  is another measure on  $\bar{\mathcal{M}}$  extending  $\mu$ , then for  $A = E \cup F$ :

$$\mu(E) = \nu(E) \leq \nu(E \cup F) \leq \nu(E \cup N) \leq \nu(E) + \nu(N) = \mu(E) + 0$$

This forces  $\nu(A) = \mu(E) = \bar{\mu}(A)$ . □

The measure  $\bar{\mu}$  in the above theorem is called the **completion of  $\mu$** , and  $\bar{\mathcal{M}}$  is called the **completion of  $\mathcal{M}$  with respect to  $\mu$** .

## Outer Measures

In this section, we develop the tools used to construct measures. To motivate the ideas, it is useful to recall the procedure used in calculus to define the area of a bounded region  $E$  in the plane  $\mathbb{R}^2$ .

### Motivation: Area in the Plane

One typically draws a grid of rectangles in the plane and approximates the area of  $E$  from two sides:

- **Inner Area:** Approximated from below by the sum of the areas of the rectangles in the grid that are *subsets* of  $E$ .
- **Outer Area:** Approximated from above by the sum of the areas of the rectangles in the grid that *intersect*  $E$ .

The limits of these approximations as the grid is taken finer and finer give the “inner area” and “outer area” of  $E$ . If they are equal, their common value is the “area” of  $E$ .

The key idea here is that of **outer area**. If  $R$  is a large rectangle containing  $E$ , the inner area of  $E$  is simply the area of  $R$  minus the outer area of  $R \setminus E$ . Thus, the notion of outer area is the fundamental concept we generalize to abstract spaces.

### Definition of Outer Measure

The abstract generalization of the notion of outer area is as follows:

**Definition 24.** An **outer measure** on a nonempty set  $X$  is a function  $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$  that satisfies the following three properties:

1.  $\mu^*(\emptyset) = 0$ .
2. **Monotonicity:**  $\mu^*(A) \leq \mu^*(B)$  if  $A \subset B$ .
3. **Countable Subadditivity:**  $\mu^*(\bigcup_1^\infty A_j) \leq \sum_1^\infty \mu^*(A_j)$  for any sequence  $\{A_j\} \subset \mathcal{P}(X)$ .

**Remark.** Note that the domain of an outer measure is the entire power set  $\mathcal{P}(X)$ , but it does not require countable additivity-only subadditivity.

**Proposition** (Proposition 1.10). Let  $\mathcal{E} \subset \mathcal{P}(X)$  and  $\rho : \mathcal{E} \rightarrow [0, \infty]$  be such that  $\emptyset \in \mathcal{E}$ ,  $X \in \mathcal{E}$ , and  $\rho(\emptyset) = 0$ . For any  $A \subset X$ , define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E} \text{ and } A \subset \bigcup_{j=1}^{\infty} E_j \right\}$$

Then  $\mu^*$  is an outer measure.

*Proof.* We verify the three axioms of an outer measure step-by-step.

**Step 1: The definition of  $\mu^*$  makes sense.**

For any  $A \subset X$ , we must ensure that the set over which we take the infimum is non-empty. Since  $X \in \mathcal{E}$  and  $A \subset X$ , we can choose the sequence  $E_1 = X$  and  $E_j = \emptyset$  for all  $j > 1$ . This sequence covers  $A$ , so the infimum is well-defined.

**Step 2:  $\mu^*(\emptyset) = 0$ .**

Since  $\emptyset \in \mathcal{E}$  and  $\rho(\emptyset) = 0$ , we can cover the empty set using  $E_j = \emptyset$  for all  $j$ . Then  $\sum \rho(E_j) = 0$ . Since  $\mu^*$  is non-negative and the infimum must be  $\leq 0$ , we have  $\mu^*(\emptyset) = 0$ .

**Step 3: Monotonicity.**

Suppose  $A \subset B$ .

- Let  $\mathcal{C}_A$  be the collection of all sequences  $\{E_j\} \subset \mathcal{E}$  that cover  $A$ , and  $\mathcal{C}_B$  be those that cover  $B$ .
- If a sequence  $\{E_j\}$  covers  $B$ , it automatically covers  $A$  because  $A \subset B$ . Therefore,  $\mathcal{C}_B \subset \mathcal{C}_A$ .
- Taking the infimum over a larger set ( $\mathcal{C}_A$ ) results in a smaller or equal value. Thus,  $\inf(\mathcal{C}_A) \leq \inf(\mathcal{C}_B)$ , which means  $\mu^*(A) \leq \mu^*(B)$ .

**Step 4: Countable Subadditivity.**

Suppose  $\{A_j\}_{j=1}^\infty \subset \mathcal{P}(X)$  and let  $\epsilon > 0$ .

1. By the property of the infimum, for each  $j$ ,  $\mu^*(A_j) + \epsilon 2^{-j}$  is not a lower bound for the sums. Thus, there exists a sequence  $\{E_j^k\}_{k=1}^\infty \subset \mathcal{E}$  such that  $A_j \subset \bigcup_{k=1}^\infty E_j^k$  and:

$$\sum_{k=1}^\infty \rho(E_j^k) \leq \mu^*(A_j) + \epsilon 2^{-j}$$

2. Let  $A = \bigcup_{j=1}^\infty A_j$ . Then  $A \subset \bigcup_{j,k=1}^\infty E_j^k$ . This double union is a countable union of elements in  $\mathcal{E}$ , so it is a valid candidate for the infimum defining  $\mu^*(A)$ .
3. By the definition of  $\mu^*$  as the *infimum*, we have:

$$\mu^*(A) \leq \sum_{j,k} \rho(E_j^k) = \sum_j \left( \sum_k \rho(E_j^k) \right)$$

4. Substituting the inequality from step (1):

$$\mu^*(A) \leq \sum_j (\mu^*(A_j) + \epsilon 2^{-j}) = \sum_j \mu^*(A_j) + \epsilon \sum_j 2^{-j} = \sum_j \mu^*(A_j) + \epsilon$$

Since  $\epsilon$  is arbitrary, we conclude  $\mu^*(A) \leq \sum_j \mu^*(A_j)$ . □

**Remark** (Observation: Construction of Lebesgue Outer Measure). *We can apply the general result of above Proposition to  $\mathbb{R}^n$  to define a measure for arbitrary sets. Let  $\mathcal{E}$  be the collection of all ***n*-boxes** in  $\mathbb{R}^n$ , where an *n*-box is a product of *n* intervals. For any box  $B$ , let  $\text{Vol}_n(B)$  denote its standard volume.*

**Definition 25.** *The ***n*-dimensional Lebesgue outer measure**  $\mu_{L,n}^*$  of an arbitrary set  $Y \subset \mathbb{R}^n$  is defined as:*

$$\mu_{L,n}^*(Y) = \inf \left\{ \sum_{j=1}^\infty \text{Vol}_n(A_j) : A_j \text{ are } n\text{-boxes with } Y \subset \bigcup_{j=1}^\infty A_j \right\}$$

*By Proposition above, since  $\emptyset$  is a box with volume 0 and  $\mathbb{R}^n$  can be covered by boxes,  $\mu_{L,n}^*$  is a valid outer measure on  $\mathcal{P}(\mathbb{R}^n)$ . This gives us a concrete way to measure the "outer size" of any set in Euclidean space using simple geometric building blocks.*



## The Carathéodory Condition

The fundamental step that leads from outer measures to measures is as follows. If  $\mu^*$  is an outer measure on  $X$ , a set  $A \subset X$  is called  $\mu^*$ -**measurable** if it satisfies the **Carathéodory condition**:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset X.$$

### Key Observations for Students:

- **Subadditivity is Given:** The inequality  $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$  holds for any  $A$  and  $E$  by the definition of an outer measure.
- **The Proof Strategy:** To prove that  $A$  is  $\mu^*$ -measurable, it is sufficient to prove only the reverse inequality:

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

- **Handling Infinite Measure:** The reverse inequality is trivial if  $\mu^*(E) = \infty$ .
- **Simplified Criterion:** Therefore,  $A$  is  $\mu^*$ -measurable if and only if:

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset X \text{ such that } \mu^*(E) < \infty.$$

**Theorem 4** (Carathéodory's Theorem). *If  $\mu^*$  is an outer measure on  $X$ , the collection  $\mathcal{M}$  of  $\mu^*$ -measurable sets is a  $\sigma$ -algebra, and the restriction of  $\mu^*$  to  $\mathcal{M}$  is a complete measure.*

*Proof.* Let  $\mu^*$  be an outer measure on  $X$  and  $\mathcal{M}$  be the collection of sets  $A \subset X$  satisfying the Carathéodory condition:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset X.$$

### Step 1: $\mathcal{M}$ is closed under complements.

The definition of the Carathéodory condition is perfectly symmetric with respect to  $A$  and  $A^c$ . Since  $(A^c)^c = A$ , the equation for  $A^c$  is identical to the equation for  $A$ . Thus,  $A \in \mathcal{M} \implies A^c \in \mathcal{M}$ .

### Step 2: $\mathcal{M}$ is an algebra and $\mu^*$ is finitely additive.

Suppose  $A, B \in \mathcal{M}$ . To show  $A \cup B \in \mathcal{M}$ , let  $E \subset X$  be any test set.

1. Since  $A \in \mathcal{M}$ , we split  $E$  using  $A$ :

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

2. Since  $B \in \mathcal{M}$ , we split the pieces  $(E \cap A)$  and  $(E \cap A^c)$  using  $B$ :

$$\mu^*(E \cap A) = \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c)$$

$$\mu^*(E \cap A^c) = \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)$$

3. Substitute these back into the original equation for  $\mu^*(E)$ :

$$\mu^*(E) = \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)$$

4. Note that  $A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$ . By the subadditivity of  $\mu^*$ :

$$\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B)$$

5. Since  $A^c \cap B^c = (A \cup B)^c$ , substituting the inequality from (4) into (3) yields:

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

This proves  $A \cup B \in \mathcal{M}$ . If  $A, B$  are disjoint, setting  $E = A \cup B$  in the split  $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$  gives  $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$ , showing finite additivity.

### Step 3: $\mathcal{M}$ is a $\sigma$ -algebra.

Let  $\{A_j\}_1^\infty$  be disjoint sets in  $\mathcal{M}$ . Let  $B_n = \bigcup_1^n A_j$  and  $B = \bigcup_1^\infty A_j$ .

1. By finite additivity and induction:  $\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j)$ .
2. Since  $B_n \in \mathcal{M}$ , we have  $\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c)$ .
3. By monotonicity,  $B^c \subset B_n^c$ , so  $\mu^*(E \cap B_n^c) \geq \mu^*(E \cap B^c)$ . Thus:

$$\mu^*(E) \geq \sum_{j=1}^n \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$$

4. Taking  $n \rightarrow \infty$ :  $\mu^*(E) \geq \sum_{j=1}^{\infty} \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$ .
5. By countable subadditivity:  $\sum_{j=1}^{\infty} \mu^*(E \cap A_j) \geq \mu^*(\bigcup (E \cap A_j)) = \mu^*(E \cap B)$ .
6. Therefore:  $\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$ , which proves  $B \in \mathcal{M}$ .

**Step 4:  $\mu^*$  is countably additive on  $\mathcal{M}$ .**

Setting  $E = B$  in the inequality  $\mu^*(E) \geq \sum \mu^*(E \cap A_j) + \mu^*(E \cap B^c)$  gives:

$$\mu^*(B) \geq \sum_{j=1}^{\infty} \mu^*(A_j) + 0$$

Since  $\mu^*(B) \leq \sum \mu^*(A_j)$  by subadditivity, we have equality.

**Step 5: The measure is complete.**

If  $\mu^*(A) = 0$ , then for any  $E \subset X$ ,  $\mu^*(E \cap A) = 0$  by monotonicity. Since  $\mu^*(E) \geq \mu^*(E \cap A^c)$  also by monotonicity, we have:

$$\mu^*(E) \geq 0 + \mu^*(E \cap A^c) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

This satisfies the Carathéodory condition, so  $A \in \mathcal{M}$ . □